Every convex body in $\mathbb{R}^3$ can be covered by 13 smaller homothetic copies

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Abstract

Every convex body in $\mathbb{R}^3$ is covered by 13 translates of its interior, so Hadwiger's covering number satisfies $H_3 \le 13$. The previous bound was 14 (Prymak). We keep Prymak's framework and add one ingredient: a minimal circumscribed parallelotope forces a chord of length at least 0.569 (per coordinate) in a face-diagonal direction for every axis. One of the 13 covering sets uses this chord. The proof is completed by an exact linear-programming certificate with 7,561,583 leaves, checked by two separately written exact checkers. The reduction from the certificate to the theorem is formalised in Lean 4. We also correct two points in the $H_3 \le 14$ argument: the symmetry-reduced region used there is not a fundamental domain, and its Lemma 2.4(ii) fails at some box corners.

1 Introduction

For a convex body $K \subset \mathbb{R}^n$ let $C(K, \operatorname{int} K)$ be the least number of translates of $\operatorname{int} K$ covering K, and $H_n = \max_K C(K, \operatorname{int} K)$. The conjecture of Levi, Hadwiger, Gohberg and Markus is $H_n = 2^n$. In dimension 3 the bounds 20 (Lassak [5]), 16 (Papadoperakis [1]) and 14 (Prymak [2]) are known; see [6] for a survey and [3, 4] for related work. Prymak notes that 14 is the limit of his method: at the central configuration there are 14 points no two of which lie in one translate of his covering body. We remove this obstruction.

Theorem 1. *Every convex body $K \subset \mathbb{R}^3$ is covered by 13 translates of $\operatorname{int} K$. Consequently K is covered by 13 sets $x_i + rK$ with $0 < r < 1$, and $H_3 \le 13$.*

2 Framework

After an affine map, $C = [0,1]^3$ is a parallelotope of minimal volume containing K. One can choose contact points $q_{i0} \in K \cap \{x_i = 0\}$ with $q_{i1} = q_{i0} + e_i \in K$ (Papadoperakis, Lemma 3(a)). They are the columns of

$$A_p = \begin{pmatrix} 0 & 1 & p_3 & p_3 & p_5 & p_5 \\ p_1 & p_1 & 0 & 1 & p_6 & p_6 \\ p_2 & p_2 & p_4 & p_4 & 0 & 1 \end{pmatrix}, \qquad V_p = \{A_p e_1, \dots, A_p e_6\}, \qquad O_p = \operatorname{conv} V_p \subseteq K,$$

for some $p \in [0,1]^6$. Let E be the edge skeleton of C.

Lemma 2 (Papadoperakis, Lemma 4)**.** *Suppose $E \cup V_p \subseteq \bigcup_{i \le m} (v_i + X_i)$, where every $X_i \subseteq \operatorname{int} K$. Then $C(K, \operatorname{int} K) \le m$.*

We prove both lemmas directly (first variation of the box volume for Lemma 3(a); for each $x \in \partial K$ the cone $x - \operatorname{cone}(K - x)$ meets $E \cup V_p$, which gives illumination directions). Both proofs are formalised.

3 The long-chord lemma

Lemma 3. *Let $[0,1]^3$ be a minimal-volume parallelotope containing K. For each axis i, with $\{j,k\}$ the other two, there is $s \in \{\pm 1\}$ such that K contains a segment of vector $\tau(e_j + se_k)$, where $\tau = \frac{1}{2} + \frac{\sqrt{2}-1}{6} = 0.5690\ldots$.*

Proof. Let $D = K - K$, so $\pm e_l \in D \subseteq [-1,1]^3$. Take $i = 1$. For $u_1 = e_1$, $u_2 = (a,1,1)$, $u_3 = (b,1,-1)$ we have $|\det U| = 2$, and the parallelotope cut out by the three slabs of K has volume $h_D(u_1)h_D(u_2)h_D(u_3)/2$. Minimality gives $h_D(a,1,1)\,h_D(b,1,-1) \geq 2$ for all a, b, so, say, $\inf_a h_D(a,1,1) \geq \sqrt{2}$. Let $(0,t,t) \in \partial D$ with outer normal $n = (a', 1+d, 1-d)$, $h_D(n) = 2t$. Then $2t \geq 1 + |d|$ and $\sqrt{2} \leq h_D(n) + h_D(0,-d,d) \leq 2t + 2|d|$, so $2t \geq 1 + (\sqrt{2}-1)/3$. □

With $w = e_j + se_k$, $\lambda = 7/10$ and $t = 569/1000 < \tau$: if $[z, z+tw] \subseteq K$ then $\lambda z + (1-\lambda)\operatorname{int} O_p + \lambda t[0,w] \subseteq \operatorname{int} K$. The position z only shifts this set.

4 Boxes and strict leaves

For a box $P \subset [0,1]^6$ let $Q_P = \bigcap_v O_v$ over the vertices v of P. Prymak's Lemma 2.4(i) gives $Q_P \subseteq O_p$ for $p \in P$, hence $\operatorname{int} Q_P \subseteq \operatorname{int} K$. A *plain leaf* is a box with 13 translation vectors such that 13 translates of $\operatorname{int} Q_P$ cover E and the six rectangles $\{A_p e_f : p \in P\}$, in a fixed combinatorial pattern (eight vertex translates, five face translates, breakpoints on edges). A *chord leaf* for axis i has, for each sign $s = \pm 1$, such a cover in which one translate is replaced by $T + \mu\lambda t w_s + (1-\lambda)\operatorname{int} Q_P$, $\mu \in [0,1]$. All memberships are *strict* linear inequalities against the exact H-representation of Q_P. By Lemmas 2 and 3, if p lies in a leaf then $C(K, \operatorname{int} K) \leq 13$.

Prymak instead uses Lemma 2.4(ii), that some translate of Q_P lies in $\operatorname{int} O_p$. This fails for $P = \prod[a_i, a_i + \frac{1}{160}]$, $a = (0, 0, \frac{63}{160}, \frac{5}{8}, 0, \frac{57}{80})$, at the corner $p = a + \frac{1}{160}(1,1,0,0,1,0)$: Q_P meets facets of O_p with inner normals $n_1 = (0, 80, 9040)$, $n_2 = (18080, -160, 0)$, $n_3 = (-480, 0, -496)$, $n_4 = (3680, -4960, 0)$, and $\frac{215760}{113} n_1 + \frac{104377}{113} n_2 + 34800\, n_3 + n_4 = 0$. So every translate of Q_P has a point outside $\operatorname{int} O_p$. Strict leaf inequalities make Lemma 2.4(ii) unnecessary.

5 Symmetry

The cube symmetries act on p by signed coordinate permutations. Let $s_x = p_3 + p_5$, $s_y = p_1 + p_6$, $s_z = p_2 + p_4$. Prymak's region $D = \{p_1, p_2, p_3 \leq \frac{1}{2},\ s_z \leq s_y \leq s_x\}$ is not a fundamental domain: $p = (\frac{459}{500}, \frac{1481}{5000}, \frac{1689}{10000}, \frac{9529}{10000}, \frac{9359}{10000}, \frac{3079}{10000})$ has no image in D or in the union of the start boxes certified in [2]; about 2.7% of random p behave this way. The reason is that a reflection sends $s_i \mapsto 2 - s_i$, so permutations can order $|s_i - 1|$ but not s_i. The region $D^* = \{p_1, p_2, p_3 \leq \frac{1}{2},\ |s_z - 1| \leq |s_y - 1| \leq |s_x - 1|\}$ meets every orbit.

We do not certify D^*. Symmetries map the 1/10 grid of $[0,1]^6$ to itself, so it suffices that every orbit of 1/10 boxes contains a certified start box. The 39,380 start boxes of Prymak's region miss 570 orbits; we certify one box of each. Given K, a symmetry g moves its configuration into a certified box, and the leaf is applied to gK.

6 The certificate and its checks

- 39,950 start boxes of side 1/10, bisected on the longest side to side at least 1/640; 7,561,583 leaves. The 999 chord leaves all lie in the 8 start boxes around $(\frac{1}{2})^6$.
- Solutions are rational with denominator 10^9, found by floating-point LPs and stored per leaf.
- Checker 1 (`certcheck_strict.py`): exact integer arithmetic, strict inequalities, tiling of every start box; all leaves pass. A non-strict variant, cross-checked against a rational reference on 1% of the leaves, also passes. A separate script checks orbit coverage of the 10^6 grid boxes.
- Checker 2 (`hcheck.py`): written from the specification only; ignores the stored structures and requires the exact intervals of the 13 translates on each edge to cover it; all leaves pass strictly, all start boxes are tiled, and every grid orbit contains a certified box.
- Lean 4: Lemmas 3(a) and 4 of Papadoperakis, existence of the minimal box, Lemma 3, Lemma 2.4(i), leaf soundness, the symmetry reduction and the passage to homothets are proved without `sorry`. The only assumptions beyond Lean's standard axioms are the certificate (every certified start box is tiled by leaves satisfying the strict leaf predicate) and one `native_decide` computation (orbit coverage of the grid).

7 Remarks

The two corrections apply to the proof of $H_3 \leq 14$ in [2]. The 570 extension boxes are certified by plain leaves, which lie inside Prymak's framework, and the strict-containment argument replaces Lemma 2.4(ii) for every leaf that satisfies its inequalities strictly. We have not re-checked Prymak's certificate under the strict test. Lemma 3 is elementary and may be known; we did not find it in the literature.

Data

- Code, checkers, Lean project and certificate hashes: `https://github.com/LeeMarshall1113/ai-math-results`, folder `results/hadwiger-h3`; archived as doi:10.5281/zenodo.23748157 (all versions).
- Certificate data (2.2 GB): doi:10.5281/zenodo.23853716. SHA-256 of the main archive: `8929c5b3599230d0d63`

AI use

This work was carried out with AI assistance (Claude, Anthropic), directed by the author. The certificate was checked by two separately written exact checkers, the reduction is formalised in Lean, and the result was attacked by adversarial AI referees. It has not been peer reviewed. The author is responsible for all content.

References

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